

scripts/compute-api-source-hash.sh

The **single source of truth** for "what lava-api-go source codebase does the on-device API embed (liblavaapi.so) contain?".

What it does

Emits a bare 64-hex sha256 on stdout (no label) over the EXACT set of Go source files that transitively compile into the c-shared .so:

- every non-test .go under lava-api-go/cmd/lavaapi-cshared/
- every non-test .go under lava-api-go/internal/
- lava-api-go/go.mod
- lava-api-go/go.sum

*_test.go files are excluded (they do not link into the .so). The other cmd/entrypoints (lava-api-go, healthprobe, lava-migrate) are excluded (not linked into the embed).

Determinism contract

1. File list sorted with LC_ALL=C sort (byte order, locale-stable).
2. Per-file digest = sha256("<module-relative-path>\0" + <file bytes>), so a rename, move, or content edit each change the digest.
3. Final digest = sha256(stream of ordered per-file digests), so reordering is impossible and the value reproduces across runs / hosts / shells / checkout locations.

Usage

```
scripts/compute-api-source-hash.sh          # prints the 64-hex hash
```

Exit 0 on success; exit 2 if lava-api-go or its source set is missing.

Who calls it

- `lava-api-go/scripts/build-cshared.sh` — injects the hash into `internal/version.SourceHash` via `-ldflags -X` and writes the committed manifest `core/apiengine/src/main/resources/api-source.hash`.
- `scripts/check-api-app-sync.sh` — the CI gate; recomputes + compares to the manifest.
- `app/build.gradle.kts` + `api-app/build.gradle.kts` — bake the hash into `BuildConfig.LAVA_API_SOURCE_HASH` at config time.

Because all four call the SAME function, there is exactly one definition of "the embed's source hash"; drift cannot be hidden. See `docs/ON_DEVICE_API.md §5A`.

Constitutional notes

§6.R (no hardcoded values — only repo-relative source paths). §6.T.2 / §11.4.67 (bounded, read-only, no host mutation). `bash -n clean`.